UML CLASS DIAGRAM REPORT

Hospital Management System

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1. Introduction

This report explains the UML class diagram designed for the Hospital Management System. The diagram presents the main classes, their attributes, operations, and relationships. It is a design model for an academic assignment rather than a deployed hospital application.

2. Classes in the Diagram

Patient
Attributes: patientID, fullName, dob, gender, contactInfo. Operations: register(), updateDetails(), viewAppointments(), viewMedicalHistory(). The class represents a person receiving care and the patient's basic details.

Doctor
Attributes: doctorID, fullName, specialization, contactInfo, salary. Operations shown in the diagram include diagnose(), prescribeMedication(), scheduleAppointment(), and viewSchedule(). The class represents a doctor who treats patients and manages appointment-related tasks.

Appointment
Attributes: appointmentID, date, time, status. Operations: schedule(), reschedule(), cancel(), and checkin(). The class represents a scheduled visit and its current status.

MedicalRecord
Attributes: recordID, diagnosis, treatmentPlan, notes, creationDate. Operations: createRecord(), updateRecord(), and viewRecord(). The class represents the clinical information recorded for a patient.

Bill
Attributes: billID, totalAmount, paymentStatus, issueDate. Operations: generateBill(), processPayment(), and viewInvoice(). The class represents the amount due and the status of payment.

3. Relationships Between Classes

Patient — Doctor (is treated by): The diagram connects Patient and Doctor using the relationship label “is treated by.” The multiplicity labels shown near this line should be read directly from the submitted diagram.

Patient — Appointment (requests): The Patient class is connected to Appointment with the label “Patient requests,” representing a patient requesting an appointment.

Patient — MedicalRecord (owns): The diagram shows that a Patient owns MedicalRecord information.

Patient — MedicalRecord (updates/views): A second labelled connection indicates updates/views between Patient and MedicalRecord as shown in the diagram.

Doctor — Appointment: Doctor and Appointment are connected, indicating the doctor’s involvement in appointment management.

Doctor — Bill (generates): The diagram shows a “Generates” relationship between Doctor and Bill.

Appointment — Bill (generates / pays): The diagram labels the Appointment–Bill area with “Generates” and “Pays.” These labels are reproduced from the drawing; confirm the exact intended direction and multiplicities against the final diagram.

4. Multiplicity

Multiplicity indicates how many instances of one class may be associated with an instance of another class. Common UML labels include 1 (exactly one), 0..1 (zero or one), 0..* (zero or many), and 1..* (one or many). The diagram includes multiplicity labels on its associations. Their exact meaning depends on the label placed at each end of each line, so they should be checked against the final exported diagram before submission.

5. Attributes and Operations

Attributes describe the information stored by a class, such as IDs, names, dates, status, and contact details. Operations describe actions associated with a class, such as registering a patient, scheduling an appointment, updating a medical record, or processing a payment. In UML notation, parentheses after an operation name indicate a method; a return type may follow a colon.

6. Design Observations

- The diagram contains five main classes: Patient, Doctor, Appointment, MedicalRecord, and Bill.
- The classes cover patient information, doctor information, appointment handling, medical records, and billing.
- Relationship names communicate the intended interactions between the classes.
- Before final submission, verify that every association line has a clear direction where needed and that multiplicities are placed at the correct ends.
- Keep class, attribute, and operation names consistent across the diagram and both reports.

7. Conclusion

The UML class diagram provides a structural view of the proposed Hospital Management System. It identifies the main classes, their data, their operations, and the relationships needed to represent basic hospital workflows. Together, the diagram and this report communicate the design of the system and provide a basis for later implementation if required.

Submission note: Add the required group members' names and student IDs if requested by the lecturer. This report follows the visible class names and labels in the diagram; review the multiplicities and association directions in the final drawing before submitting.